AI/ML Engineer Study Plan — Phase 1 (Weeks 1 to 5)

Target: Software & Data Foundations | **Curriculum**: Video-by-Video Breakdown with Notes & Practice Tasks

Pacing & Strict Video Scope: Study pace is ~2 to 2.5 hrs/day across 4-5 active days per week with pre-budgeted festive buffers (Diwali Nov 5-8 & weddings). Only the exact videos required for each week are listed. Do not jump forward into later videos prematurely.

Week 1: Python Core Essentials, Input/Output & Conditionals

Dates & Pacing: Nov 2 – Nov 8, 2026 (Diwali on Nov 8; Nov 6-8 Light Festive Buffer)
Primary Goal: Write clean Python programs with variables, data types, user input, math operators, conditionals, and loops.
English Resource: Harvard CS50P: Week 0 (Functions/Variables) & Week 1 (Conditionals) | **Math**: Khan Academy India (Hindi): Fractions, Decimals, Order of Operations (www.khanacademy.org/math/hindi)
Notes & Guides: CodeWithHarry Python Ch 1-2 Notes & Harvard CS50P Notes Week 0-1
Week's Concrete Deliverable: **Build Number Guessing Game CLI & Complete Exercises 1–10 from Starter Pack**

Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
Day 1	**Day #1: Intro to Programming & Python**	What is programming, interpreter vs compiler, Python 3 installation, VS Code setup.	Watch Video
Day 1	**Day #2: Some Amazing Python Programs**	Ecosystem overview: AI, ML, Web APIs, Automation. What Python can do.	Watch Video
Day 1	**Day #3: Modules and Pip in Python**	Built-in modules vs external modules, pip install, terminal commands.	Watch Video
Day 1	**Day #4: Our First Python Program**	print('Hello World'), code execution flow, indentation rules.	Watch Video
Day 2	**Day #5: Comments & Escape Sequences**	Single/multi-line comments, \n, \t, print sep=' ' and end='\n' parameters.	Watch Video
Day 2	**Day #6: Variables & Data Types**	Variables, int, float, str, bool, complex, type() function, type inspection.	Watch Video
Day 2	**Day #7 & #8: Operators & Calc Project**	Arithmetic operators (+, -, *, /, //, %, **). Build basic calculator CLI.	Watch Video
Day 3	**Day #9: Typecasting in Python**	Explicit vs implicit type conversion, converting str to int/float.	Watch Video
Day 3	**Day #10: Taking User Input**	input() function, handling user text and casting to numbers immediately.	Watch Video
Day 4	**Day #14: If Else Conditionals**	Comparison operators (==, !=, >, <, >=, <=), if, elif, else indentation logic.	Watch Video
Day 4	**Day #17: For Loops in Python**	for item in sequence, iterating strings, range(start, stop, step) function.	Watch Video
Day 5	**Day #18: While Loops in Python**	while condition, loop counters, incrementing, avoiding infinite loops.	Watch Video

Week 2: Data Structures, Functions & Git Fundamentals

Dates & Pacing: Nov 9 – Nov 15, 2026 (Bhai Dooj on Nov 10; Weekend Wedding Buffer)
Primary Goal: Master Python data collections (Lists, Tuples, Dictionaries, Sets), reusable functions, file handling, and Git version control.
English Resource: Harvard CS50P: Week 2 (Loops & Collections) & Week 6 (File I/O) | **Math**: Khan Academy: Coordinate Planes, Graphing Equations & Linear Expressions
Notes & Guides: CodeWithHarry Notes Ch 3, 4, 5, 8 & Pro Git Book (git-scm.com/book/en/v2)
Week's Concrete Deliverable: **Contact Book CLI with JSON storage (Add, List, Search, Delete) committed to GitHub**

Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
Day 8	**Day #11 & #12: Strings & Slicing**	String indexing, negative indexing, slice notation [start:stop:step], len().	Watch Video
Day 8	**Day #13: String Methods in Python**	upper(), lower(), strip(), replace(), split(), count(), find(), startswith().	Watch Video
Day 9	**Day #20 & #21: Functions & Arguments**	def, parameters vs arguments, default values, return statement, *args/**kwargs.	Watch Video
Day 10	**Day #22 & #23: Lists & List Methods**	List indexing, mutable behavior, append(), insert(), extend(), pop(), sort().	Watch Video
Day 11	**Day #24 & #25: Tuples in Python**	Tuple immutability, tuple methods, packing/unpacking, converting to list.	Watch Video
Day 11	**Day #31 & #32: Sets & Set Operations**	Unordered unique elements, union(), intersection(), difference(), symmetric_diff.	Watch Video
Day 12	**Day #33 & #34: Dictionaries & Methods**	Key-value mappings, dict[key] vs dict.get(), keys(), values(), items(), update().	Watch Video
Day 13	**Git & GitHub Crash Course (Hinglish)**	git init, git add, git commit, git status, git log, creating repo on GitHub, push.	Watch Video

Week 3: Object-Oriented Programming (OOP) & Practical Python

Dates & Pacing: Nov 16 – Nov 22, 2026 (Wedding / Reception Weekend Buffer)
Primary Goal: Write modular, professional code using Classes, Constructors, Inheritance, Exception Handling, and File I/O.
English Resource: Harvard CS50P: Week 8 (Object-Oriented Programming) | **Math**: Khan Academy: Statistical Measures (Mean, Median, Mode, Variance Intuition)
Notes & Guides: CodeWithHarry Notes Ch 9, 10, 11 & Official Python Docs on Classes
Week's Concrete Deliverable: **Refactor Contact Book into clean OOP classes with Custom Exceptions and Logging**

Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
Day 15	**Day #36 & #37: Exception Handling**	try, except, else, finally blocks, ValueError, ZeroDivisionError handling.	Watch Video
Day 15	**Day #38: Raising Custom Errors**	raise keyword, inheriting from Exception, defining meaningful error messages.	Watch Video
Day 16	**Day #49 & #50: File I/O in Python**	open(filename, mode), 'r', 'w', 'a', with open(...) as f context managers, readlines().	Watch Video
Day 17	**Day #56 & #57: Intro to OOPs & Classes**	Classes as blueprints, objects as instances, attributes, self parameter.	Watch Video
Day 17	**Day #58: Constructors (__init__)**	__init__ dunder method, initializing object state, instance attributes.	Watch Video
Day 18	**Day #60: Getters and Setters**	@property decorator, encapsulation, protecting internal state.	Watch Video
Day 19	**Day #61: Inheritance in Python**	Single inheritance, child class inheriting parent methods, code reuse.	Watch Video
Day 19	**Day #72 & #73: super() & Dunder Methods**	super() call, __str__, __repr__, __len__ magic methods.	Watch Video

Week 4: Relational Databases & SQL for Data/ML

Dates & Pacing: Nov 23 – Nov 29, 2026 (Event / Social Buffer Day Included)
Primary Goal: Model data relationships, create schemas, and write complex analytical queries with joins and aggregations.
English Resource: Kaggle: Intro to SQL & Advanced SQL Courses | **Math**: Khan Academy: Set Theory, Venn Diagrams & Relational Logic
Notes & Guides: Kaggle SQL Tutorials & PostgreSQL Official Tutorial Docs
Week's Concrete Deliverable: **E-Commerce Database Schema (Users, Products, Orders) + 15 Analytical Queries**

Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
Day 22	**SQL Foundations: Databases & Tables**	What is RDBMS, SQLite vs PostgreSQL, tables, rows, columns, data types.	Watch Video
Day 22	**SQL CRUD: CREATE, INSERT, SELECT**	CREATE TABLE, Primary Keys, INSERT INTO, basic SELECT * FROM table.	Watch Video
Day 23	**Filtering & Sorting: WHERE, ORDER BY**	WHERE clause, comparison operators, AND, OR, NOT, IN, BETWEEN, LIKE, LIMIT.	Watch Video
Day 24	**Aggregations: GROUP BY & HAVING**	COUNT(), SUM(), AVG(), MIN(), MAX(), GROUP BY columns, filtering with HAVING.	Watch Video
Day 25	**Relational Modeling & JOINS**	Foreign keys, 1-to-many relationships, INNER JOIN, LEFT JOIN, FULL OUTER JOIN.	Watch Video
Day 26	**Subqueries & Window Functions**	Nested SELECT subqueries, Common Table Expressions (WITH CTE), ROW_NUMBER().	Watch Video

Week 5: NumPy & Pandas for Data Manipulation

Dates & Pacing: Nov 30 – Dec 6, 2026 (Catch-up / Wedding Buffer Day Included)
Primary Goal: Manipulate tabular data and arrays, clean real-world datasets, handle missing values, and prepare data for ML.
English Resource: FreeCodeCamp: Data Analysis with Python (NumPy & Pandas Full Course) | **Math**: Khan Academy: Matrix Addition, Multiplication, Dot Products & Summary Statistics
Notes & Guides: 10 Minutes to Pandas Official Guide & NumPy Quickstart
Week's Concrete Deliverable: **Clean and profile a Kaggle dataset (e.g. Telco Churn CSV) with summary tables**

Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
Day 29	**NumPy Arrays: Creation & Attributes**	np.array(), ndarray attributes (.shape, .dtype, .ndim), np.zeros, np.ones, np.arange.	Watch Video
Day 29	**NumPy Vectorization & Math**	Element-wise math (+, -, *, /), dot products (@), broadcasting rules, memory speed.	Watch Video
Day 30	**NumPy Indexing, Slicing & Masking**	2D slicing arr[row, col], boolean masking arr[arr > 5], filtering arrays without loops.	Watch Video
Day 31	**Pandas Series & DataFrames**	pd.Series, pd.DataFrame, reading CSV files (pd.read_csv), .head(), .info(), .describe().	Watch Video
Day 32	**Data Cleaning & Preprocessing**	Handling missing values (.isna(), .dropna(), .fillna()), deduplication, type casting.	Watch Video
Day 33	**Filtering & Aggregation in Pandas**	Boolean indexing df[df['age'] > 25], .groupby('category').agg(), .pivot_table().	Watch Video
Day 34	**Merging & Combining DataFrames**	pd.merge() for SQL-like joins (inner, left), pd.concat() for row/column stacking.	Watch Video